SAI RAM MADALA

Baltimore, MD, USA | +1 (240)-459-8769 | rammadala1710@gmail.com | LinkedIn

SUMMARY

AI/ML Engineer with 3+ years of experience engineering Generative AI applications, enterprise RAG pipelines, MLOps frameworks, and recommendation engines. Proficient in LangGraph, LangChain, Azure OpenAI, AWS SageMaker, vLLM, and Databricks. Demonstrated expertise optimizing LLM inference latency from 15 hours to 7 minutes, scaling multi-tenant AI infrastructure, and deploying production-grade evaluation pipelines utilizing MLflow, Ragas, and NeMo Guardrails.

TECHNICAL SKILLS

Programming And Tools: Python, SQL, Java, Scala, Bash/Shell, Git, GitHub, Agile, SDLC, A/B Testing.
Generative AI & LLMs: RAG Pipelines, Hybrid Search, Cross-Encoder Rerankers, Agentic AI, LangChain, LangGraph, LlamaIndex, Azure OpenAI, Anthropic MCP, Prompt Engineering, Fine-Tuning (LoRA, QLoRA), Quantization (GGUF/AWQ), vLLM, NeMo Guardrails, Ragas.
Vector Databases & Serving: Pinecone, FAISS, ChromaDB, Milvus, Triton Inference Server, TensorRT-LLM.
Machine Learning: Scikit-learn, XGBoost, LightGBM, Classification, Regression, Clustering, Recommendation Systems, Feature Engineering, Time-Series Forecasting.
Deep Learning & NLP: TensorFlow, PyTorch, Keras, Transformers, BERT, CNN, RNN, LSTM, NER, PII Masking (Presidio).
MLOps & Observability: MLflow, LangSmith, Model Monitoring, Drift Monitoring (PSI/CSI), Evidently AI, Model Validation, Experiment Tracking.
Cloud & Data Platforms: AWS (SageMaker, Bedrock, S3, EC2), Azure (Machine Learning, Databricks, Azure OpenAI), Snowflake, Docker, Kubernetes, GitHub Actions, CI/CD.
Backend & APIs: FastAPI, Flask, REST APIs, Microservices, Apache Spark, Spark SQL, Delta Lake, Great Expectations, Pandas, NumPy, SciPy.

PROFESSIONAL EXPERIENCE

AI/ML Engineer | Synaptiq.ai | USA *July 2025 – December 2025*

- Architected multi-agent evaluation workflows using LangGraph, MLflow, Azure Databricks, LangSmith, and NeMo Guardrails to validate 1,000+ legal documents and establish model observability.
- Developed document classification workflows using Azure OpenAI, Spark SQL, and Pydantic validation schemas across 300+ secure enterprise endpoints.
- Improved enterprise RAG document extraction accuracy by 22% across 18 document categories by implementing hybrid search architectures combining BM25 keyword matching with dense vector embeddings.
- Integrated advanced cross-encoder rerankers (Cohere/BGE) into retrieval steps, reducing irrelevant chunk extraction and lowering hallucination rates.
- Optimized LLM inference execution using vLLM and parallel processing techniques, reducing total runtime from 15 hours to 7 minutes.
- Executed prompt caching, model quantization, and token reduction strategies to decrease production AI inference operational costs.
- Scaled multi-tenant AI document platforms for 450+ enterprise tenants using Azure ML, OpenAI, and automated DevOps CI/CD deployment workflows seamlessly.
- Engineered automated PII masking and prompt injection defense layers using Presidio to secure sensitive enterprise datasets in production endpoints.
- Designed asynchronous API endpoints using FastAPI to handle high-concurrency document processing requests.
- Implemented distributed Redis caching for vector embeddings and queries, cutting downstream API latency by 28%

ML Engineer | Dell Technologies | India *May 2022 – July 2024*

- Developed Scikit-learn and Pandas machine learning models to analyze marketplace behavior across 500K+ user interaction records.
- Built Apache Spark, SQL, and Apache Airflow feature engineering flows supporting 25+ automated model training cycles.
- Configured automated data validation sequences using Great Expectations to maintain data integrity across distributed ingestion stages.
- Automated model packaging and deployment via Docker, AWS SageMaker, and Git across development and production environments.
- Configured model performance and drift monitoring using MLflow, Population Stability Index (PSI) metrics, and Evidently AI across 40+ production training cycles.
- Evaluated recommendation systems using TensorFlow, Tableau, and controlled A/B testing frameworks to improve ranking accuracy.
- Streamlined large-scale data ingestion and preprocessing workflows, reducing data processing execution time by 30%.
- Migrated legacy SQL query structures to optimized Apache Spark jobs to enhance cluster compute efficiency.
- Authored comprehensive unit testing suites using PyTest, increasing code coverage across core machine learning repositories.
- Constructed distributed log aggregation and telemetry tracing pipelines using OpenTelemetry and AWS CloudWatch to debug multi-node ML training failures.
- Optimized memory utilization in large-scale Pandas and NumPy dataframes through chunked processing and explicit data-type casting, preventing out-of-memory cluster crashes.

PROJECTS

FluCast – AI-Based Influenza Forecasting & Decision Support Platform

- Engineered FastAPI inference microservices to process 51 state-level forecasting requests through XGBoost structures, reducing API response latency by 35%.
- Integrated automated data validation checks, ensuring 99.8% data integrity prior to triggering forecasting execution workflows.
- Designed system logging and monitoring mechanisms to track real-time model accuracy metrics across production deployments.

OpAL (Open Source AI for Legal) – AI-Powered Subpoena Response Automation System

- Developed a Python-based RAG architecture using LangGraph, LlamaIndex, Sentence Transformers, and Milvus/Pinecone vector databases to automate legal compliance document evaluation.
- Implemented LoRA fine-tuned models and vector retrieval optimization techniques to achieve 93% extraction accuracy while reducing manual review effort by 40%.
- Optimized vector retrieval latency and contextual relevance by tuning HNSW indexing parameters in Milvus.
- Implemented multi-stage evaluation metrics via Ragas to benchmark model faithfulness and answer relevance against legal ground-truth datasets during automated compliance testing cycles.

EDUCATION

Master of Science in Computer Science
University of Maryland, Baltimore County (UMBC), Baltimore, MD, USA August 2024 – May 2026
GPA: 3.96 / 4.00